UPI Block-and-Debit for Solar Slot Booking

How to hold a customer's funds when they book a slot, and collect automatically on the slot date, in our solar module aggregator app.

1. Summary

The IPO mechanism you are referring to is UPI ASBA: the customer approves a **block** on their bank account, nothing leaves the account, and the amount is debited only when the event happens (allotment). NPCI generalised this as the **UPI One-Time Mandate (OTM) / Single-Block-Multiple-Debit** facility, which other merchants can use too. The idea maps directly onto slot booking:

- Customer books a slot, then approves a UPI block for the booking amount.
- Money stays in their account but cannot be spent elsewhere.
- On the slot date, our backend triggers a debit against the block (full or partial).
- If the booking is cancelled or modified, the unused block is released.

Important caveat: the block facility is not open to every merchant category. IPO and capital-market flows are pre-approved. For a solar aggregator, eligibility, per-block limits and validity windows depend on our acquiring bank/PSP and NPCI rules at the time. These must be confirmed with the PSP before we commit to the design (see section 8).

2. How the mechanism works

Step	What happens
Mandate creation	We send a block request (amount, validity, payee, purpose) through our PSP; the customer receives a collect/intent request in their UPI app
Customer approval	Customer enters UPI PIN; the issuing bank places a hold for that amount
Block active	Funds are earmarked; customer sees reduced available balance

Step	What happens
Execution	On the due date we send an execute/debit instruction for an amount up to the blocked amount
Release	Any unused part of the block is released automatically at expiry, or earlier if we send a revoke

Key properties: the debit cannot exceed the blocked amount, the debit must happen inside the validity window, and the customer can see and revoke the mandate in their UPI app.

3. Fit with slot booking

Works well when: the slot date is soon (days to a few weeks), the amount is known at booking time, and the aim is to reduce no-shows and secure commitment without taking cash upfront.

Works poorly when: the slot is far in the future, longer than the maximum block validity. In that case the block must be created later (for example, reminder and re-authorisation closer to the date) or we need a different instrument (section 7).

4. Proposed booking flow

1. Customer selects module, quantity and slot date.
2. App creates a booking in status `PENDING_BLOCK` and reserves inventory for a short hold (for example 10 minutes).
3. Backend calls the PSP to create the block mandate; the app opens the customer's UPI app (intent) or shows a collect request.
4. Customer approves with UPI PIN.
5. PSP webhook confirms the block. Booking moves to `CONFIRMED_BLOCKED`; inventory is locked.
6. A scheduler runs ahead of the slot date and sends a reminder (also required for transparency before any debit).
7. On the slot date (or on dispatch/delivery milestone), backend triggers the debit. Booking moves to `PAID`.

8. If cancelled before the cutoff, backend revokes the block. Booking moves to `CANCELLED`, block released.
9. If the block expires or the debit fails, booking moves to `PAYMENT_FAILED` and triggers a fallback (payment link, manual follow-up).

5. Booking state machine

`PENDING_BLOCK` → `CONFIRMED_BLOCKED` → `PAYMENT_DUE` → `PAID`

Off-ramps: `BLOCK_FAILED`, `CANCELLED`, `BLOCK_EXPIRED`, `PAYMENT_FAILED`, `PARTIALLY_PAID`.

6. Technical implementation

Data model (minimum)

- `bookings`: id, customer_id, slot_date, amount, status, inventory_hold_expiry
- `upi_mandates`: id, booking_id, psp_mandate_ref, blocked_amount, valid_from, valid_to, status, customer_vpa
- `mandate_events`: id, mandate_id, event_type, raw_payload, received_at (audit trail of every webhook)
- `debits`: id, mandate_id, amount, psp_txn_ref, status, attempted_at

Backend components

- PSP integration service: create block, check status, execute debit, revoke.
- Webhook handler: verify signature, deduplicate by event id, update status idempotently.
- Scheduler / job queue: pre-debit reminder, debit execution on due date, expiry sweeper, reconciliation.
- Reconciliation job: daily match of our ledger against PSP settlement reports.

Frontend

- Clear consent screen stating amount, what is blocked, the date it will be debited and how to cancel.
- UPI intent deep link on mobile, QR or collect request on desktop.
- Booking page showing mandate status and a cancel option.

Engineering rules

- Every PSP call and webhook must be idempotent (use booking id as idempotency key).
- Never assume a debit succeeded from the client; trust only the PSP webhook plus status check.
- Store the PSP reference, not card/bank details. Do not store the UPI PIN or anything sensitive.

7. Edge cases and alternatives

Edge cases

- Customer has insufficient balance at block time: block fails, booking is not confirmed.
- Customer revokes the mandate in their UPI app: webhook marks `CANCELLED_BY_CUSTOMER`, release inventory per policy and notify sales.
- Slot rescheduled: if the new date is inside validity, keep the block; otherwise revoke and recreate.
- Price change (for example module price revision): debit cannot exceed the blocked amount, so block with a small buffer or recreate the block.
- Partial delivery: debit the partial amount and release the rest.
- Bank downtime on the debit date: retry window with backoff, then fallback to payment link.

Alternatives if the block is not available or validity is too short

1. UPI AutoPay (recurring mandate) with pre-debit notification. Works for scheduled debits but does not hold funds, so it does not guarantee money on the date.
2. Token advance via a normal payment gateway, with the balance collected later. Simplest, and standard for B2B solar.
3. Bank guarantee or LC for large commercial orders. UPI limits will be too low for utility or C&I scale deals.

The practical point: UPI limits per transaction and per mandate are small relative to solar module order values. This design suits residential and small commercial orders. For large orders use RTGS/NEFT, LC or escrow.

8. Compliance and onboarding checklist

- Confirm with our PSP / acquiring bank that our merchant category is eligible for block (OTM) mandates, and obtain the current per-mandate amount limit and maximum validity.
- Merchant onboarding with NPCI-certified PSP; sandbox access and UAT sign-off.
- Customer consent text and cancellation policy approved by legal.
- Refund and dispute process (chargeback handling for UPI).
- Data protection: privacy notice for payment data, retention period for mandate records (DPDP Act obligations).
- GST invoice generated at debit time, tied to the payment reference.

9. Rollout plan

Phase	Scope
1. Feasibility (1 to 2 weeks)	Talk to 2 or 3 PSPs, confirm eligibility, limits, validity, fees
2. Sandbox build (2 to 3 weeks)	Booking states, PSP integration, webhooks, scheduler
3. Pilot (2 to 4 weeks)	Limited customers, small orders, manual reconciliation as backup
4. Scale	Full rollout, monitoring dashboards, ops runbook

10. Open questions for you

- Typical order value and how far ahead slots are booked? (This decides whether UPI block is feasible at all.)
- Is the debit on the slot date, or on delivery/dispatch?
- Who are the buyers: individuals, installers, or commercial customers?
- Which PSP or payment gateway are you already using?